MCPulse — Architecture on Amazon Web Services

MCP Server Monitoring & Performance Analytics Platform | MCA Semester III, Cloud Computing | Account 462854211377 | Region us-east-1 | Deployed 2026-10-21/22

1 Deployment topology

```
Browser (any network) --HTTP 80 / HTTPS 443--> [ Amazon VPC vpc-09e2f7d67772fdfea ]
  +-------------------------------------------------------------------------------------------+
  | EC2 i-0059c1813dd3d1ada t3.small / Amazon Linux 2023 54.161.184.121 (us-east-1b) |
  | +-------------------------------------------------------------------------+ |
  | | Docker Compose: nginx :80 (host networking) -> Node.js/Express API :3001 | |
  | | (static React bundle + reverse proxy) (loopback only, 10 s scheduler) | |
  | +-------------------------------------------------------------------------+ |
  | security group mcpulse-web-sg sg-085f461886a0676f7 (80/443 public, 22 one IP) |
  +-------------------------------------------------------------------------------------------+
  | TLS 5432 (security group to security group)
  v
  RDS mcpulse-db PostgreSQL 16.13 db.t3.micro 20 GB gp3 Single-AZ PubliclyAccessible = false
  mcpulse-db.cyinxjr3lgjz.us-east-1.rds.amazonaws.com:5432 sg mcpulse-db-sg sg-0d74bb805b76e8e0e
  tables: mcp_servers · monitoring_checks · incidents · mcp_tools · schema_migrations
```

2 Data flow

```
Monitoring cycle (every 10 s)
scheduler -> SELECT servers FROM RDS
  -> HTTPS POST "initialize" (JSON-RPC 2025-03-26)
  -> GitHub / Google Drive / Brave /
    Cloudflare / Hugging Face
  -> latency + status + sanitised error
  -> INSERT monitoring_checks
  -> 3 failures: INSERT incidents (open)
  -> 1 success : UPDATE incidents (resolved)

User request
browser -> nginx -> /api/* -> Express API
  -> auth key + SSRF guard
  -> TLS to RDS (CA + hostname verified)
  -> JSON to the dashboard

Observability
API + host -> CloudWatch Logs (4 groups, 7 d)
host -> CloudWatch Metrics (CWAgent mem/disk)
```

3 Services and why

Service	Role	Reason
Amazon VPC (default)	network isolation, 6 AZs	the database has no path out of the VPC
Amazon EC2 t3.small	nginx + API + scheduler	long running stateful service; t3.micro cannot build the images
Amazon RDS PostgreSQL 16	persistent store, 20 GB gp3	managed, patched, isolated, encrypted at rest
Security group (web)	80/443 public, 22 one IP	port 3001 deliberately not exposed
Security group (db)	5432 from web SG only	database reachable only from the app tier
CloudWatch Logs	app + host logs, 7 day retention	diagnostics with bounded cost
CloudWatch Metrics	mem / disk of the instance	capacity trends without SSH
IAM LabRole	present, not attachable	iam:PassRole denied by the lab - documented finding
Amazon S3	not used	CSV is produced from the database; no needless cost

4 Security controls (all verified) 5 Cost 6 Evidence

Database isolation - PubliclyAccessible = false; security group accepts 5432 only from the web tier; RDS refuses unencrypted connections.
TLS - certificate chain *and* hostname verified against the Amazon RDS CA bundle; the ssl* URL parameters are stripped so the verified configuration cannot be overridden.
Attack surface - the API is published on 127.0.0.1:3001 only; nginx is the single public entry point; every state changing request needs the administrative key.
SSRF - loopback, private, carrier grade NAT, link local and cloud metadata destinations are rejected before any outbound request (169.254.169.254 -> HTTP 400).
Secrets - database password, SSH key and AWS credentials live outside the repository; the tree is scanned clean.

EC2 t3.small $0.021 / h
RDS db.t3.micro $0.017 / h
20 GB gp3 $2.30 / month
Public IPv4 $0.005 / h
~ $0.18 per 4 h session
no ALB, no NAT gateway, no Multi-AZ, no S3

5 monitored providers · 1333 recorded checks · 5 incidents · 4 tools discovered (hf_fs, hf_whoami, hub_repo_details, hub_repo_search) · 3 applied migrations · 18 AWS console screenshots in **AWS SS/** · machine readable evidence in **docs/DEPLOYMENT_STATUS.md** · reproducible with **scripts/aws/{audit,deploy,resume,stop,allowlist}.sh**. verified live known limits: no instance profile (iam:PassRole), CloudWatch log events not delivered in this account, Ubuntu AMI not visible